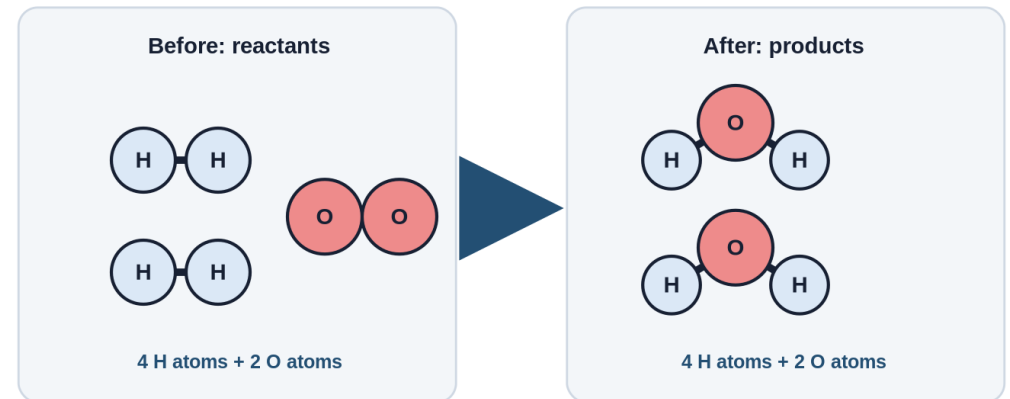


Intro Molecules

Models, chemical change, and temperature change

Chemical change model: atoms rearrange



Learning target

I can make and draw a model of molecules showing the composition before and after a chemical change as well as what happens when temperature changes.

What a molecule model shows

A model is a useful simplification - not a photograph.

Model key: circles stand for atoms



Hydrogen atom

same color = same element



Oxygen atom

same color = same element



Carbon atom

same color = same element



Nitrogen atom

same color = same element

A molecule is two or more atoms bonded together.

Atom

One particle of an element. In our model, one circle = one atom.

Molecule

Two or more atoms bonded together. Bonded atoms move as a unit.

Composition

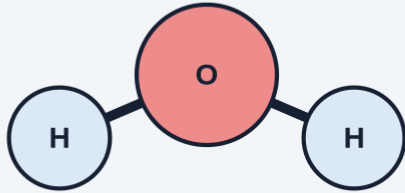
The types and number of atoms in a molecule.

Read the code in a model

Color tells the element. Connections tell what is bonded.

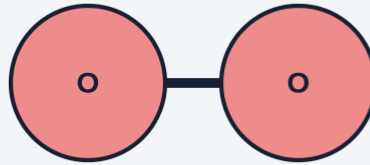
Read the code in molecule models

H₂O water molecule



2 hydrogen + 1 oxygen

O₂ oxygen molecule



2 oxygen atoms bonded

Water molecule

Composition: 2 hydrogen atoms + 1 oxygen atom

Oxygen molecule

Composition: 2 oxygen atoms bonded together

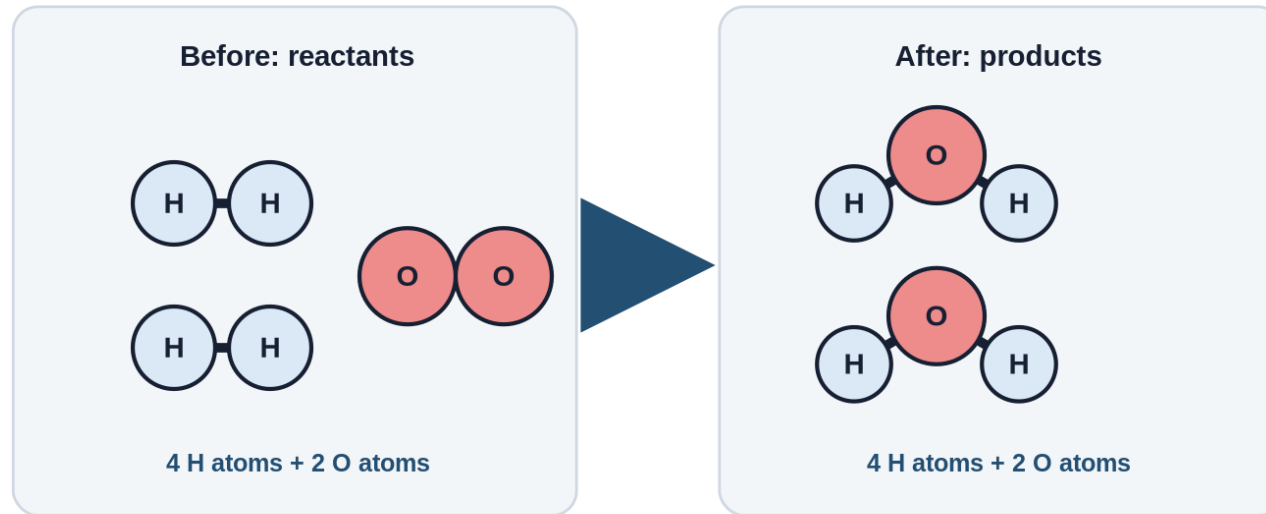
Model reading move

Say what stays the same: atom colors/counts. Say what changes: partners, spacing, or motion.

Before and after: count atoms

Chemical models must keep track of matter.

Chemical change model: atoms rearrange



Count before

4 H atoms and 2 O atoms

Count after

4 H atoms and 2 O atoms

Claim

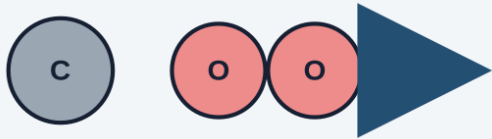
The atoms were rearranged into new molecules.
The atoms were not created or destroyed.

Chemical change = new particle arrangement

Same atoms. New partners. New substances.

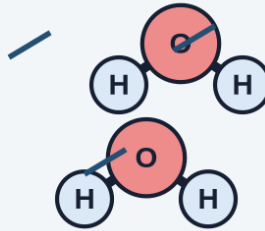
Two kinds of model changes

Chemical change



Bonds/partners change

Temperature change



Speed/spacing changes

Evidence in a model

Bonds or partners are different after the change.

Not enough

Molecules moving faster or spreading out is not by itself a chemical change.

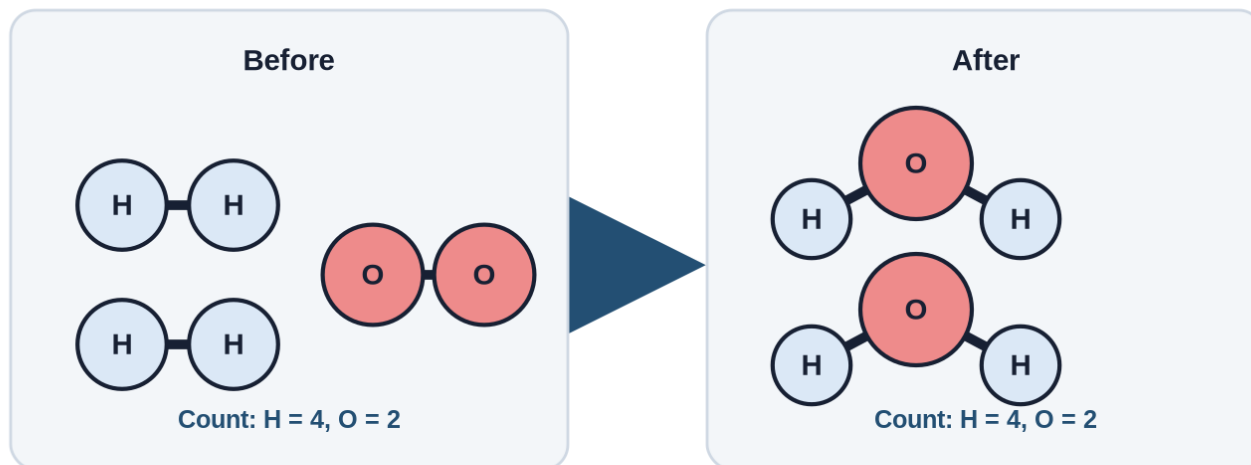
Model check

Can you account for every atom before and after?

Teacher model: how to draw a chemical change

Use the same atom key on both sides.

Draw-check-count process



Step 1: Key

Choose one color/letter for each atom type.

Step 2: Before

Draw the starting molecules. Keep atoms bonded correctly.

Step 3: After

Redraw the same atoms in new molecules.

Step 4: Count

Check each atom type before and after.

Draw-check-count

Use the same colors, draw new molecules, then count each atom type on both sides.

Formative check 1: Which model works?

Silent think -> partner talk -> whole class.

Model A

Before: 2 red atoms + 2 white atoms
After: 1 red atom + 2 white atoms

Does it conserve atoms?

Model B

Before: 2 red atoms + 4 white atoms
After: 2 red atoms + 4 white atoms

Does it conserve atoms?

Explain

Sentence frame:

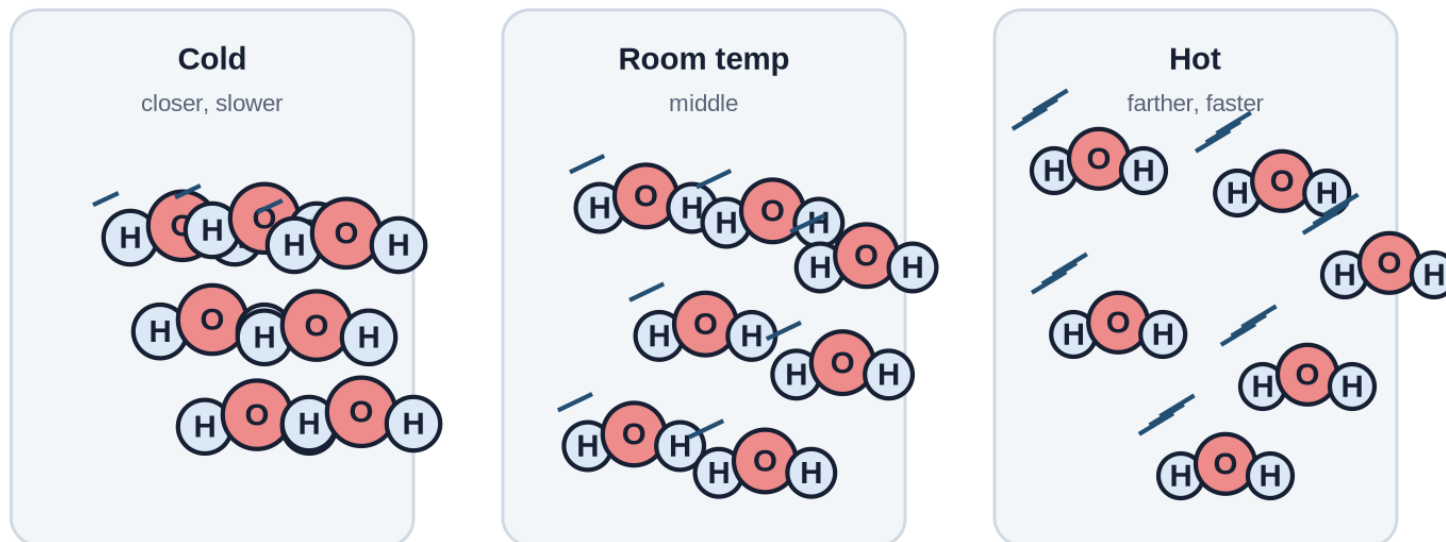
"I think ___ works because the model shows ___ before and ___ after."

Teacher look-for: students justify with atom counts, not with picture neatness.

Temperature changes molecular motion

Heating and cooling change how molecules move.

Temperature change model: same molecules, different motion



Heating does not turn water molecules into different molecules.

Heating

Molecules move faster and tend to be farther apart.

Cooling

Molecules move slower and tend to be closer together.

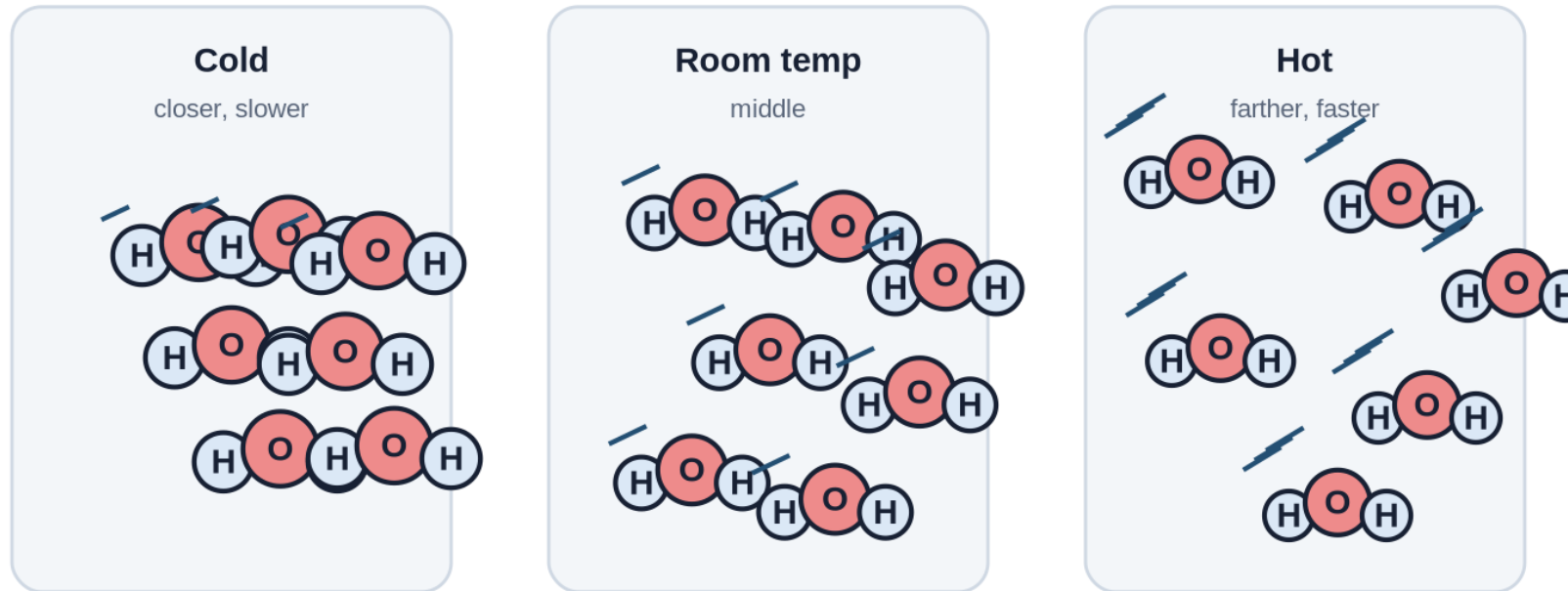
Important

The water molecules are still H₂O molecules. This is a motion/spacing change.

How to draw temperature in a model

Do not change the molecule formula. Change spacing and motion lines.

Temperature change model: same molecules, different motion



Heating does not turn water molecules into different molecules.

Cold

closer dots, fewer motion lines

Room temperature

middle spacing and motion

Hot

farther dots, more motion lines

Compare the two model stories

Ask: did the composition change?

Chemical change

What changes in the model?

- Atoms have new partners
- New molecules form
- Composition of molecules changes

What stays the same?

- Total atom count for each element

Temperature change

What changes in the model?

- Molecules move faster/slower
- Spacing changes

What stays the same?

- Molecule identity/composition

Key question: Are the atoms rearranged into different molecules, or are the same molecules just moving differently?

Formative check 2: Draw the model

Individual draw -> pair check.

Student task

Use the atom key below. Draw a before-and-after model for:

2 H_2 molecules + 1 O_2 molecule $\rightarrow$ $2 \text{ H}_2\text{O}$ molecules

Then count H and O atoms before and after.

Pair check questions

1. Did you use the same colors on both sides?
2. Did every atom from before appear after?
3. Did your after model show new molecules, not just moved-apart molecules?

H = white circle O = red circle

Exit ticket

One model. Three claims.

1. Draw

Draw a molecule-level before-and-after model for a chemical change using at least two atom types.

2. Count

Write the atom counts before and after.

3. Explain

One sentence: "This shows a chemical change because ____."

Bonus connection

How would your drawing change if the temperature increased but no chemical change happened?